Athari — Final Project Summary

Owner: Baraka Mitigoa **Version:** `v1.0.0` **Date:** 17 October 2026 **Repository:** https://github.com/bmitigoa/athari

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1. What Athari is

Athari (Swahili for "impact") is a Drupal 11 portal for a Kenyan NGO. It lets the organisation:

1. keep a record of its **projects**, the **sites** where they run, and its **partners**;
2. collect **field reports** from staff in the counties, check them through a **review workflow** and publish only approved ones;
3. show its **impact** (people reached, shillings spent) as totals, charts and a map;
4. accept **donations by M-Pesa** to a chosen project;
5. share its data with other systems through a **read-only public API**.

All money is in **Kenyan shillings (KES)**. The geography is Kenya's 8 former provinces and 47 counties. The organisations, projects and people in the demo data are fictional.

It was built in five days as a portfolio project for NGO IT and information-systems roles in Kenya.

athari Impact you can see

Home Projects Impact Map About Log In

NGO PROJECT & IMPACT PORTAL · KENYA

Tracking change, one community at a time

Athari shows where our projects work, who they reach and what every shilling achieves, from the first field report to the final result.

Explore projects | See our impact | Donate with M-Pesa

Impact at a glance

6	KES 138.5K	39,360	KES 59.9M
ACTIVE PROJECTS	DONATIONS RECEIVED	BENEFICIARIES REACHED	FUNDS SPENT

Featured projects

Tana River Flood Response & Recovery
Sector: Protection
County: Tana River
Budget: KES 31,000,000

Kakamega Dairy Value Chain Strengthening
Sector: Livelihoods
County: Kakamega
Budget: KES 21,000,000

Kibera Maternal & Child Health Outreach
Sector: Health
County: Nairobi
Budget: KES 18,500,000

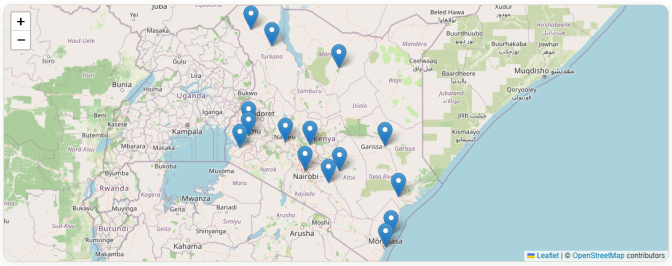
Garissa and Homa Bay Learning Recovery Programme
Sector: Education
County: Garissa
Budget: KES 12,000,000

Makueni Sand Dams & Savings Groups
Sector: Livelihoods
County: Makueni
Budget: KES 9,850,000

Nyeri Coffee Outgrowers Training
Sector: Livelihoods
County: Nyeri
Budget: KES 6,400,000

See all projects

Project map

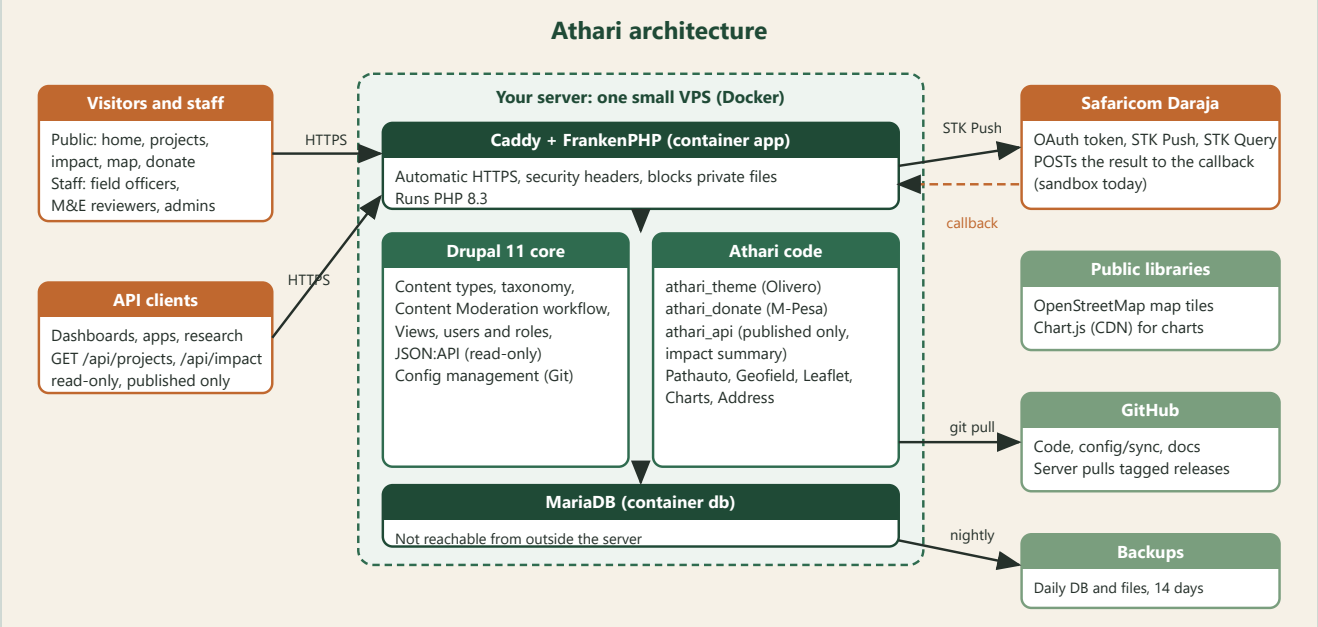


Help us reach more communities

Donate to a project of your choice by M-Pesa. Every contribution is tracked from the first payment to the field report that shows what it achieved.

Donate with M-Pesa

2. Architecture



Layer	What	Notes
Content management	**Drupal 11.4** on PHP 8.3	Content types, taxonomy, users, workflow, views, theming
Data	**MariaDB 11.8**	Not reachable from outside the server in production
Web server	**FrankenPHP (Caddy)**	Automatic HTTPS, security headers, blocks private files
Custom code	`athari_donate`, `athari_api`, `athari_theme`	The only code written for the project (everything else is configuration or standard modules)
Configuration	`config/sync` in Git	The whole site is rebuilt by `drush site:install --existing-config`
Environments	DDEV locally; Docker Compose bundle for servers	Development tools only in the `dev` configuration split
External	Safaricom **Daraja** (M-Pesa), OpenStreetMap tiles, Chart.js CDN	Daraja is sandbox only

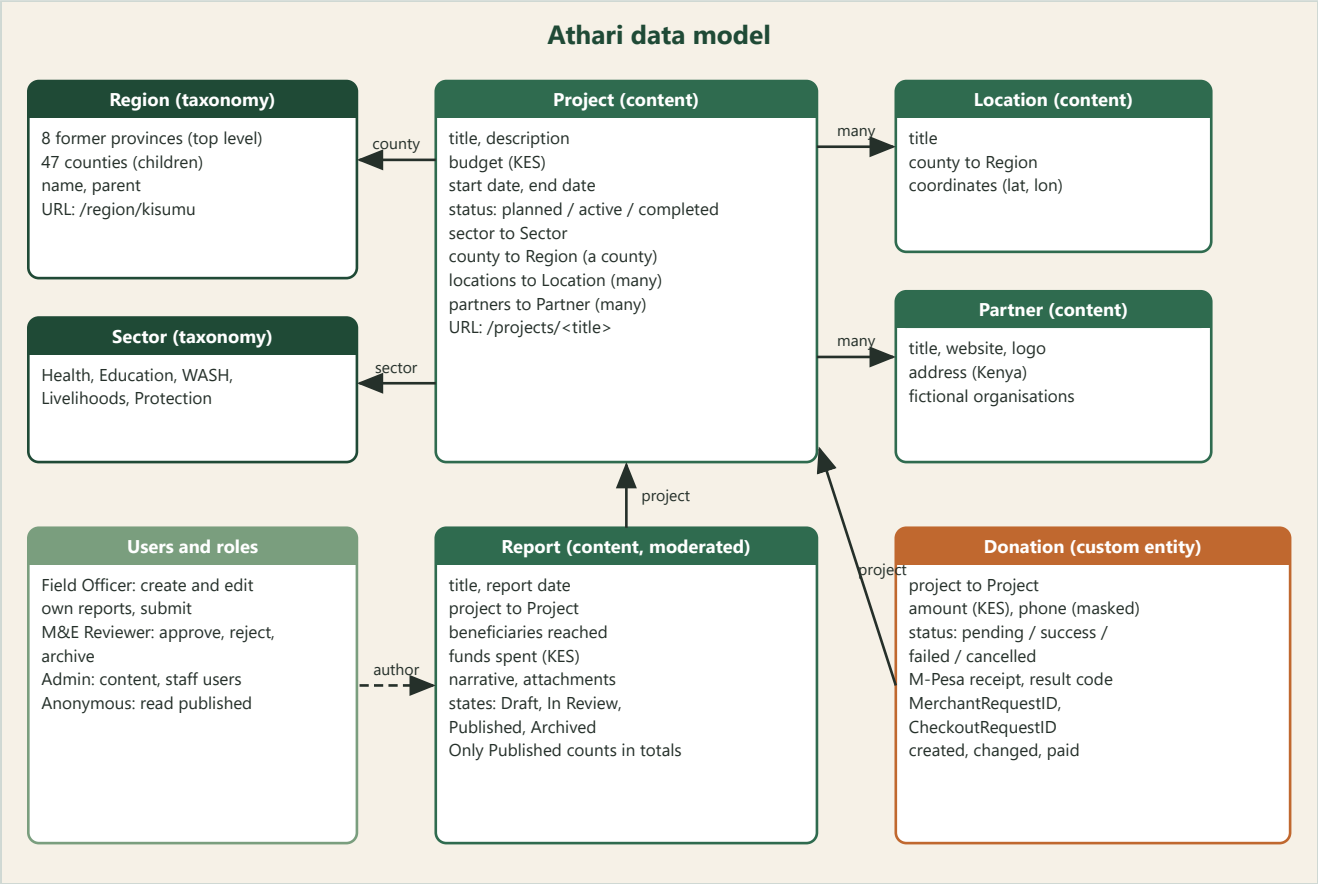
How the pieces connect

- **Content model → everything else.** Projects, locations, partners and reports are *content types*. The views (listing, dashboard, map, review queue), the API and the donation pages all read these same records, so a report approved by a reviewer changes the dashboard, the API and the homepage figures at once.
- **Workflow → trust.** Only **Published** reports count in totals and in the API. A report is Draft, In Review, Published or Archived, and what a person may do depends on their role.
- **Configuration → repeatability.** Every content type, field, view, role, workflow and setting is exported to `config/sync`. A new machine or server gets the same site from Git, plus the Kenya sample data from scripts.

- **Donations → projects.** A donation is a custom *entity* that points at a project. Success updates the project's totals, which are shown on the project page and the homepage.
- **Secrets stay out of Git.** Daraja keys, the callback token and the database password live in `settings.local.php` (git-ignored, or generated on the server).

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3. Data model



Item	Type	Main fields
Sector	Taxonomy	Health, Education, WASH, Livelihoods, Protection
Region	Taxonomy (hierarchy)	8 provinces (parents), 47 counties (children)
Project	Content type	title, description, budget (KES), start and end date, status (Planned, Active, Completed), sector, county, locations, partners
Location	Content type	title, county, coordinates (Geofield latitude and longitude)
Partner	Content type	title, logo, website, address
Report	Content type (moderated)	project, report date, beneficiaries reached, funds spent (KES), narrative, attachments
Donation	Custom entity	project, amount (KES), phone, status (pending, success, failed, cancelled), M-Pesa receipt, result code and text, request ids, timestamps
Page	Content type	The About page

4. Every feature

4.1 Content and navigation (Day 1)

- The five content types and two vocabularies above, with entity references between them (a project points at a sector, a county, locations and partners; a report points at a project).
- **Pathauto** makes readable addresses such as `/projects/<title>` and `/region/<county>`.
- **Address** is used for partner addresses, **Geofield** for coordinates, **Admin Toolbar** for faster admin menus.

4.2 Roles and the review workflow (Day 2)

Role	Can do
Anonymous	Read published content, view the dashboard, map and API, donate
Field Officer	Create reports, edit their own, submit them for review
M&E Reviewer	See the review queue, approve (publish), send back to draft, archive
Admin	Manage content, create Field Officers and Reviewers, see donations
Administrator	Technical owner: modules, configuration, permissions

The workflow `report_review` has four states (Draft, In Review, Published, Archived) and transitions between them, enforced by **Content Moderation**.

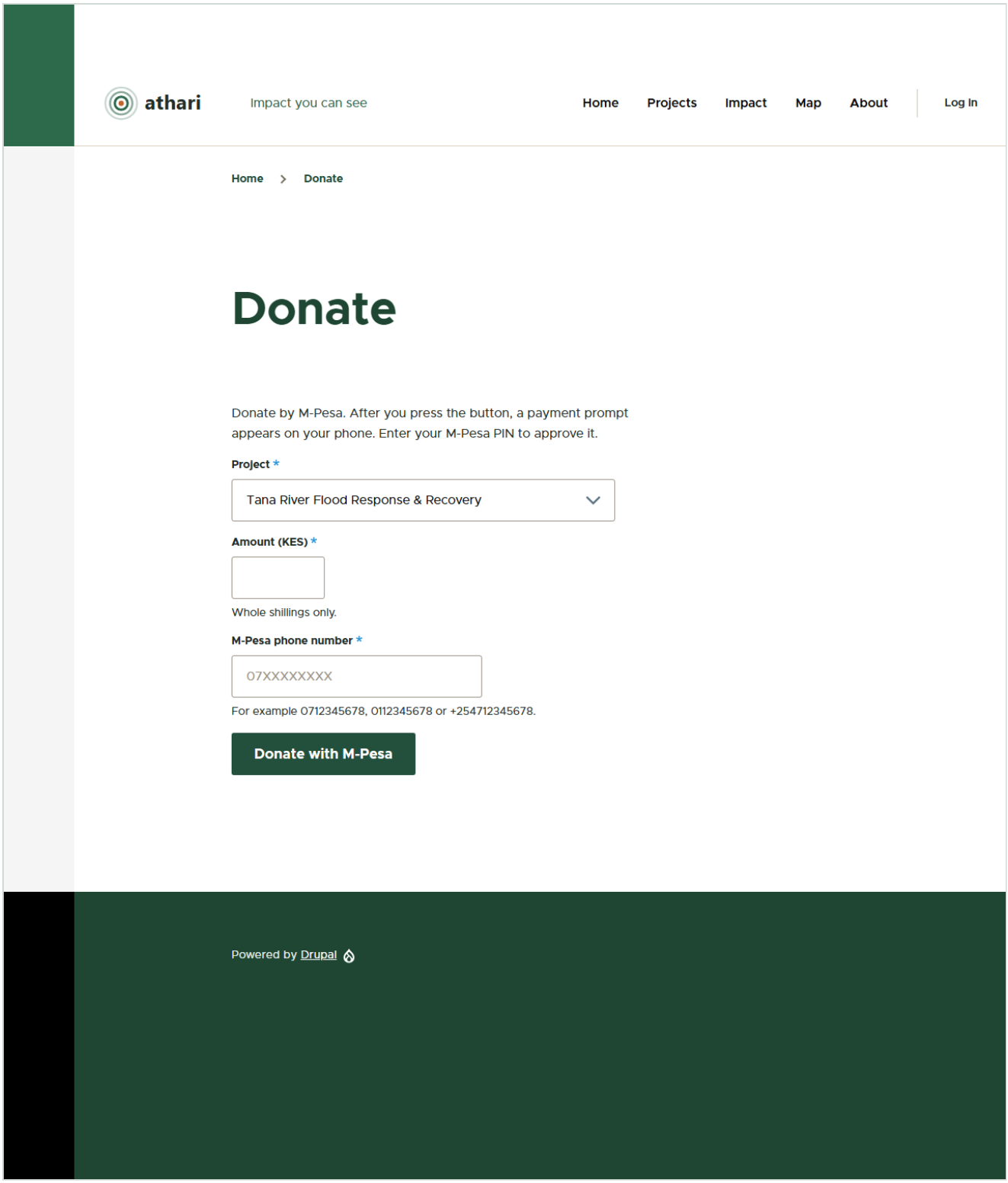
4.3 Dashboard, map, theme (Day 3)

- **Projects list** with filters for sector, county and status, as a responsive table.
- **Impact dashboard** with totals (people reached, funds spent, projects, published reports) using Views aggregation, and charts by sector and by region (Charts with Chart.js).
- **Project map** using Leaflet and Geofield on OpenStreetMap, centred on Kenya.
- **Review queue** for reviewers.
- `athari_theme`, a subtheme of Olivero with a green and earth palette, logo, hero section, stat cards, featured projects and a homepage map. It also works on a phone.

4.4 Donations by M-Pesa (Day 4, module `athari_donate`)

Part	Role
`DarajaClient`	Gets and caches the OAuth token, sends the STK Push, and queries the status of a payment
`DonationForm`	Project, amount (minimum KES 1) and phone number, validated as `2547XXXXXXXX` by `PhoneNumber`
`Donation` entity	One record per attempt, with its status and the receipt
`CallbackController`	Receives Safaricom's result at `/athari/mpesa/callback/{token}`; the path contains a secret token, processing is idempotent and rate-limited

Part	Role
`StatusController`	Donor status page and a JSON endpoint that the page polls; falls back to the STK Query if the callback does not arrive
`DonationManager`	Creates donations, applies results, expires old pending ones on cron
`SettingsForm`	`/admin/config/athari/donate`; values set in `settings.local.php` show as read-only
`PaymentProviderInterface`	The extension point for Airtel Money Kenya or cards later
Views	Admin donations report and totals per project, shown on the project and home pages



4.5 Read-only public API (Day 5, module `athari_api` and JSON:API)

- Six public resources under `/api`: projects, locations, partners, reports (published only), sectors, regions (provinces and counties), plus `/api/impact`.
- Never exposed: donations, unpublished reports, users, e-mail addresses, credentials, revisions and internal flags. Verified by 37 anonymous checks.
- Fail-closed: a resource is public only if it is listed. Writes are refused, CORS allows only GET.
- Documented with real responses in `docs/api.md`.

4.6 Security and operations (Day 5)

- Least-privilege roles checked against the design, Admin separated from Administrator, Role Delegation for staff creation.
- Errors hidden, trusted host patterns, private files folder, update status, cron.
- Development tools only in the `dev` configuration split; the production image is built without them.
- No secret in the repository or its history (8 checks).
- Deployment bundle with HTTPS, firewall, backups, updates (see section 6).

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5. Quality evidence

Check	Result
PHPUnit unit and kernel tests (all custom modules)	57 tests, 197 assertions, OK
PHPCS, Drupal and DrupalPractice standards	clean
Page and permission checks (Days 2 to 4, rerun on Kenya data)	all passing (`day-2/3/4-verification-kenya.txt` , `kenya-verification.txt`)
Anonymous API verification	37 of 37
Security configuration checks	46 of 46
HTTP security checks	39 of 39 (DDEV), 48 of 48 (live mode on the local production stack)
Secrets and history	8 of 8; `composer audit` clean
Production installer on a local Docker stack	passed
`linux/arm64` image	builds and starts
Quick-start from a fresh clone of the GitHub repository	passed: install from configuration, Kenya demo data, 9 pages answer 200, 37 API checks, 4 demo logins (see the Day 5 report)

The raw results are saved in `docs/reports/` .

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6. Running, deploying and maintaining it

Run it locally

```
 git clone https://github.com/bmitigoa/athari.git
cd athari
ddev start
ddev composer install
ddev drush site:install --existing-config -y --account-name=admin --account-pass=admin
ddev drush php:script scripts/kenya-00-sectors.php
ddev drush php:script scripts/kenya-01-regions.php
ddev drush php:script scripts/day2-03-users.php
ddev drush php:script scripts/kenya-03-seed.php
ddev drush php:script scripts/day3-06-menu-about.php
ddev drush cr
ddev launch
```

Logins (local only): `admin/admin`, `ngoadmin/ngoadmin123`, `fieldofficer/fieldofficer123`, `reviewer/reviewer123`.

Deploy it

Athari is **deployment-ready, not hosted**. One command on a fresh Ubuntu 24.04 server installs everything (Docker, firewall, HTTPS, Drupal from configuration, demo data, cron, nightly backups). Steps for Oracle Cloud Always Free (Ampere A1) and any VPS are in `docs/deployment.md`.

Maintain it

Task	How
Change configuration	Change it in the admin UI on the local site, then `ddev drush cex -y`, commit, deploy, `drush deploy` (runs updates, imports configuration, clears caches)
Update Drupal and modules	`ddev composer update "drupal/*" --with-all-dependencies`, `ddev drush updb -y`, `ddev drush cex -y`, run the tests, commit. Watch the *Available updates* report
Back up	Server: `deploy/backup.sh` (daily, 14 days). Local: `ddev export-db`
Add staff	Log in as Admin, *People, Add user*, give the Field Officer or M&E Reviewer role
Check security	`scripts/day5-04-security-check.php`, `day5-05-security-http.ps1`, `day5-02-verify-api.ps1`, `composer audit`
Rotate Daraja keys	Edit `settings.local.php` (or `/opt/athari/secrets/settings.local.php` on a server) and clear caches. Never commit them

Extend it

Goal	Where
A new payment provider (Airtel Money Kenya, cards)	Implement `PaymentProviderInterface` in `athari_donate` and choose it in the donation form
A new public API resource	Add it in JSON:API Extras (the API is fail-closed, so nothing appears until listed) and add checks to `day5-02-verify-api.ps1`
A new report field	Add the field in the UI, export configuration, and add it to the API field list if it is public
Swahili pages	Language and Content Translation are enabled; add the language and translate
A new sector or county	Add the taxonomy term in the admin UI

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7. How the five days fit together

Day	Delivered
1	DDEV site, Git, content model, taxonomies, Pathauto
2	Roles, report workflow, permissions, sample data, permission tests
3	Views, impact dashboard with charts, map, theme, homepage
4	M-Pesa donations module tested against the Daraja sandbox
Pivot	Moved the project to Kenya: KES, 47 counties, Kenyan sample data, M-Pesa as the only provider
5	Public API, security pass, deployment bundle, README, diagrams, reports, release `v1.0.0`

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8. Honest limits

- **Not hosted.** It runs locally; the deployment bundle has been exercised on a local Docker stack, not on a public server.
- **M-Pesa is sandbox only.** Production requires merchant onboarding with Safaricom. The sandbox cannot approve payments, so successful payments are shown through seeded data and a simulated callback in Safaricom's documented format.
- **Demo data is fictional**, and the local demo passwords are deliberately simple.
- **Recommended before real use:** two-factor login, a password policy, a self-hosted chart library, an organisation e-mail address in the site settings, an API rate limit, and a Safaricom IP allow-list for the callback.

9. Roadmap

Airtel Money Kenya and card payments, two-factor login, Swahili translations, host the portal publicly, apply for Safaricom production access.